

AN2DL - Second Homework Report NMD

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1. Introduction

This project tackles a *semantic segmentation problem* using **custom-trained deep learning models** to classify pixels in 64×128 grayscale images of Martian terrain into five distinct surface classes. The key steps in this study include:

- **Dataset Analysis & Preprocessing:** Analyzing the dataset to understand its structure and distribution, followed by preprocessing to normalize and prepare images and masks for training.
- **Model Design & Training:** Designing and training a neural network architecture from scratch, without using pre-trained models, to address the specific requirements of the task.
- **Performance Evaluation & Validation:** Evaluating the model's performance using metrics tailored for semantic segmentation, ensuring precise class predictions for terrain classification.

2. Problem Analysis

2.1 Dataset Characteristics

During the initial exploration of the dataset, we identified an outlier (*the alien Paul*), a corrupted segmentation mask visually distinct from the other data points. To address this issue:

- We systematically compared each sample in the dataset with the corrupted mask and flagged duplicate instances.

- The flagged samples were removed from both the input images and the corresponding masks to ensure dataset integrity.

This step guaranteed the removal of unwanted noise, preventing potential biases or errors during the model training phase.

2.2 Pixel Distribution and Class Imbalance

An analysis of the dataset revealed a significant class imbalance in the pixel distribution across the five terrain classes. As shown in Figure 3, classes 1 and 3 were overrepresented, while classes 0, 2, and especially class 4 were underrepresented. This imbalance can negatively affect model performance by causing it to focus more on dominant classes and underperform on minority ones.

To address this, we implemented two strategies:

- **Dataset cleaning:** Redundant or corrupted samples were removed to improve data quality.
- **Custom loss function:** We developed a loss function that accounts for class-specific pixel frequencies, assigning higher weights to underrepresented classes and lower weights to dominant ones. This helps balance the model's attention during training.

Figure 2 shows the pixel distribution after cleaning. A comparison of Figures 3 and 2 confirms that the imbalance was not caused by corrupted images. As a result, the focus shifted to using the custom loss function to address the class imbalance during model training.

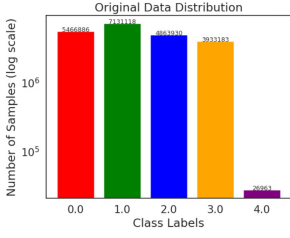


Figure 1: Original Pixel Distribution

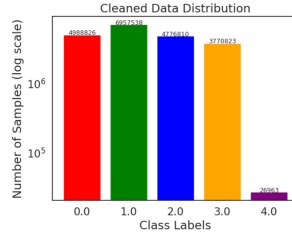


Figure 2: Cleaned Pixel Distribution

3. Data Augmentation

We employed several augmentation techniques to enhance the model’s robustness and generalization:

- **Flipping:** Both horizontal and vertical flips allow the model to learn from different orientations, making it invariant to the image’s spatial alignment. This is crucial in segmentation tasks where terrain features can appear in various orientations.
- **Rotation:** Random rotations introduce variability in image angles while preserving the spatial relationships between pixels. This transformation helps simulate changes in viewpoint, enabling the model to recognize terrain features regardless of their orientation.
- **Translation:** Translation shifts the features within the image, helping the model become invariant to the position of terrain elements. This is important since terrain features in real-world scenarios can appear in different locations within the image.

These *geometric augmentations* are particularly effective for grayscale images, where the model must rely on texture and shape rather than color.

4. Custom Metric: Mean Intersection Over Union (MIoU)

To evaluate the performance of our model, we implemented a custom version of the *Mean Intersection Over Union* (MIoU) metric.

4.1 How the Metric Operates

In Our custom version of the Mean Intersection Over Union, we excluded certain classes from the computation -most notably the background class. By ignoring the background, the metric provides a clearer assessment of the model’s ability to differentiate between terrain features such as soil, bedrock, sand, and large rocks.

4.2 Motivation for Customization

The customization of the MIoU was driven by the unique characteristics of the dataset:

- **Task-Specific Focus:** The background class dominates the dataset, which could lead to misleadingly high MIoU scores if included in the metric calculation. By excluding it, the metric focuses on the performance of the model on the less frequent but scientifically significant classes (including irrelevant classes would dilute the metric’s relevance to this goal).

4.3 Benefits of the Custom MIoU

This tailored metric offers several advantages over its standard counterpart:

- **Improved Interpretability:** By concentrating on foreground classes, the metric provides a more precise and interpretable evaluation of the model’s segmentation capabilities.
- **Enhanced Robustness:** The focus on challenging classes, which are often underrepresented, provides a more rigorous test of the model’s performance.

5. Loss Function

The performance of a semantic segmentation model is heavily dependent on the design of its loss function, as it dictates how the model learns to distinguish between different classes in the data. For this project, a custom loss function was implemented to handle the unique challenges faced.

5.1 Challenges Addressed by the Loss Function

Mars terrain segmentation presents several unique challenges:

- **Class Imbalance:** The dataset includes a significant imbalance between classes, with some (e.g., large rocks) being underrepresented.
- **Sparse Labels:** The background occupies a large proportion of the image, requiring the loss function to prioritize nonbackground classes without being overwhelmed by the background.
- **Fine Detail Preservation:** Precise segmentation of small or thin structures (e.g., small patches of sand) is crucial for scientific analysis, demanding a loss that emphasizes these features.

5.2 Structure of the Loss Function

To address these challenges, our loss function combines three complementary loss components:

- **Weighted Dice Loss (DL):** This loss is effective in handling class imbalance by directly maximizing overlap between the predicted and true segmentation masks. It applies class-specific weights inversely proportional to class frequency, ensuring that underrepresented classes are given more importance.
- **Focal Loss (FL):** Designed to address sparse labels, focal loss downweights well-classified examples and focuses the model’s learning on hard-to-classify pixels. The γ parameter controls the degree of focusing, enabling the model to better handle challenging regions.
- **Sparse Categorical Cross-Entropy (SCE):** This standard loss provides a baseline for penalizing incorrect predictions. It is computed while ignoring the background class, ensuring that the model focuses on meaningful features.

5.3 Weighted Combination of Losses

The total loss is calculated as a weighted sum of the three components:

$$\text{Total Loss} = \alpha \cdot \text{DL} + \beta \cdot \text{FL}(\gamma) + \text{SCE} \quad (1)$$

where α and β control the relative importance of the dilate and focal losses and γ is the *focusing parameter of the Focal Loss*. These weights were chosen empirically to balance the priorities of class imbalance, sparse labels, and overall precision.

6. Model Architecture and Design Choices

In this section, we detail the key design choices made in developing the model for our semantic segmentation task. The goal was to balance model complexity with effectiveness, given the constraint of training the model from scratch without pre-trained weights.

6.1 The UNet Model

For our segmentation task, we utilized the UNet architecture, a popular model known for its effectiveness in

pixel-level segmentation. The UNet consists of two primary components:

1. **Encoder (Contracting Path):** This part progressively reduces the image’s spatial dimensions through convolutional and pooling layers.
2. **Decoder (Expanding Path):** The decoder reconstructs the image’s spatial dimensions using transposed convolutions or upsampling layers.

The encoder and decoder are connected by a **bottleneck** layer, which captures the most abstract features and provides the model with global context before the decoder reconstructs the fine details. This architecture strikes a balance between preserving local features and capturing global context, making it ideal for semantic segmentation.

6.2 Model Evolution

Our initial approach involved a standard UNet with a 1024-input-channel bottleneck, but this model underperformed. We then explored a Double UNet architecture to increase model complexity, but this resulted in worse performance. Consequently, we reduced the bottleneck size to 256 input channels, which led to a significant improvement in performance.

We also tested an attention-based version of UNet, but it did not meet expectations. Adding residual connections to the model enhanced performance by improving gradient flow and enabling better feature reuse across layers. The final model uses a 1×1 convolutional output layer, with the number of filters matching the number of terrain classes. A **softmax** activation function is applied to generate class probabilities for each pixel.

- **Softmax Activation:** Normalizes the model’s output, ensuring that each pixel is assigned a probability distribution over the terrain classes.
- **Pixel-Wise Classification:** The 1×1 convolution performs independent classification for each pixel, utilizing the learned contextual features to make accurate predictions.

7. Conclusion

In this project, we tackled the challenge of semantic segmentation by implementing a UNet architecture, custom loss function to address class imbalance and sparse labels, and a tailored Mean Intersection Over Union (MIoU) metric focused on the most relevant terrain classes. These strategies significantly enhanced the model’s segmentation performance. The final model (score 0.61), represents the optimal configuration after several iterations, including adjustments to the bottleneck size and the incorporation of residual connections.

Figure 3: Best model accuracy and loss behavior

